



Islington college
(इस्लिङ्टन कलेज)

Module Code & Module Title

**CS6P05NI Project
Proposal**

5% Individual Coursework

Submission: Final Submission

AY 2025 2026

Academic Semester: Autumn Semester 2025

Credit: 30 Credit Year Long Module

Project Title: Resilio-Wellness App

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Assignment Due Date: Wednesday, December 3, 2025

Assignment Submission Date: Wednesday, December 3, 2025

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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Acknowledgement

I express my thanks to my external supervisor, Ms. Oshirya Manandhar, for her consistent guidance and support during the preparation of this proposal. Her direction helped me shape the idea and move the project in the right path.

I also thank my internal supervisor, Mr. Aadesh Tandukar, for reviewing my work and offering helpful suggestions. His feedback strengthened the proposal and improved its clarity.

I am grateful to Islington College and London Metropolitan University for providing the academic environment and resources needed to begin this project. Their support encouraged me to develop this proposal with confidence.

Abstract

This proposal presents the development plan for Resilio, a mobile application designed to support mental health and emotional wellness. The app targets users experiencing stress, anxiety, low mood, or ongoing emotional pressure in their daily lives.

Resilio provides a set of practical features including self-help tools, therapist booking, mood logging, guided relaxation sessions, and brief motivational content. The design combines input from mental health professionals with user-friendly functionality to encourage regular engagement.

The primary objectives are to help users become more aware of their emotional patterns, develop healthier daily habits, and take early action when facing psychological difficulty. The project follows a structured development timeline with defined milestones, aiming to deliver a functional product with all core features by the completion date.

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1. Introduction

1.1. Background and Context

Mental health problems are common but often ignored, especially in poorer countries. Globally, nearly one billion people live with some form of mental disorder and most never get proper help (Patel et al., 2018). Nepal is no different. The Nepal Mental Health Survey found that about 12.9 percent of adults have experienced a mental health condition at some point in their lives, with depression, anxiety and alcohol problems being the most common (Ministry of Health and Population, 2020). Around 6 percent reported having suicidal thoughts. These are not small numbers.

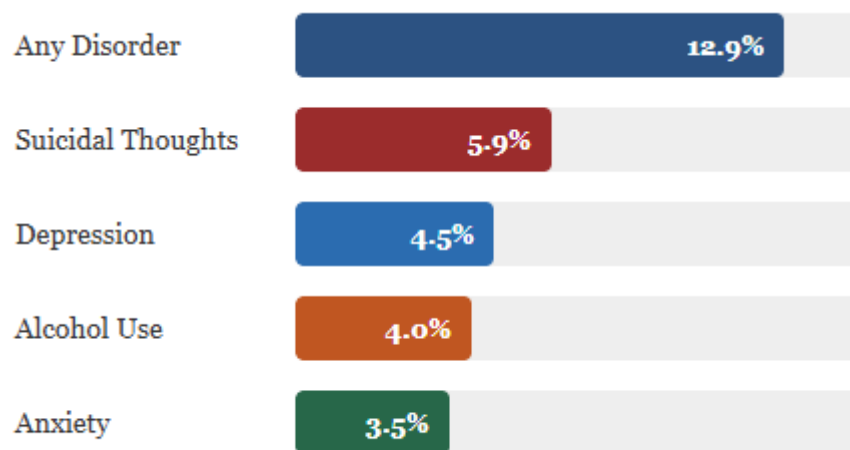


Figure 1: Prevalence of Mental Disorders in Nepal (Source: Ministry of Health and Population)

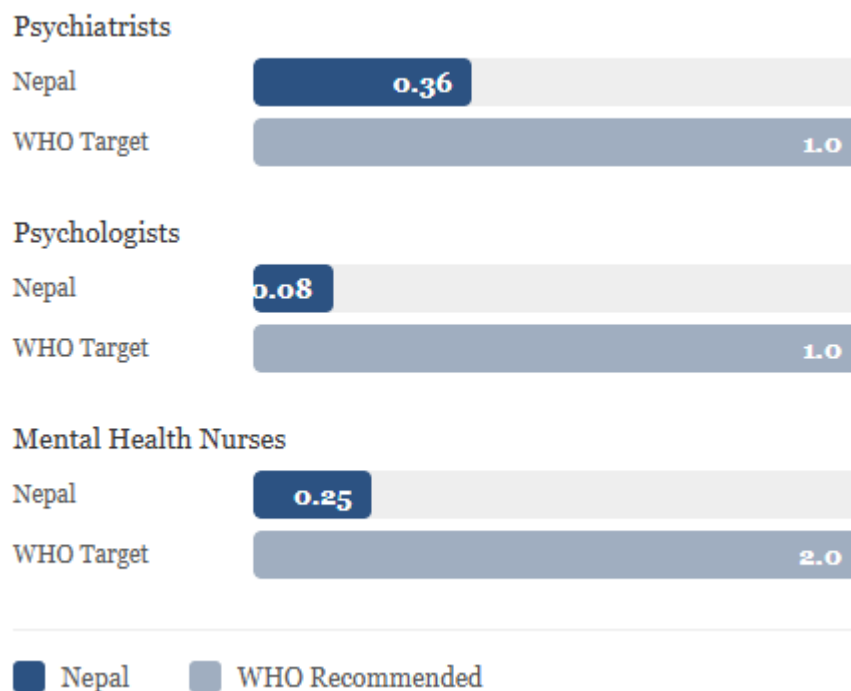


Figure 2: Mental Health Workforce per 100,000 Population (Source: World Health Organization)

[For detailed statistics please see [Appendix A: Detailed Mental Health Statistics in Nepal](#)]

1.2. Problem Domain

Existing mental health apps operate in isolation. Meditation apps, mood trackers, and therapy platforms do not integrate with each other. Research shows users abandon disconnected tools within weeks because they lack continuity between features (Bakker et al., 2016). This fragmentation is especially problematic in Nepal where scarce mental health workers need integrated digital tools to support task-sharing models of care (Patel et al., 2011). Most apps also lack long-term engagement features and require constant connectivity, making them unsuitable for Nepal's context where internet access remains inconsistent (Torous et al., 2019).

[For extended problem analysis please see [Appendix B: Extended Problem Analysis](#)]

1.3. Current Scenario

Most popular mental health apps were not built for places like Nepal. Calm and Headspace charge around \$70 per year, which is more than many Nepalis spend on healthcare in total. BetterHelp connects users to therapists but a single session cost what some families earn in a week. These apps assume you have reliable internet, a credit card, and comfort using services designed in English for Western users. They work well in New York or London, but not in Dolakha or Doti.

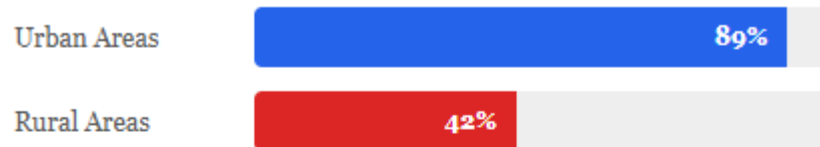
Feature	BetterHelp	Calm	Headspace	Local Apps	Resilio
Cost & Accessibility					
Monthly Cost	\$60-90	\$15	\$13	Free-\$5	Free-\$10
Local Payment (eSewa)	No	No	No	Some	Yes
Offline Mode	No	Partial	Partial	No	Yes
Nepali Language	No	No	No	Yes	Yes
Professional Support					
Licensed Therapists	Yes	No	No	No	Yes
Live Sessions	Yes	No	No	No	Yes
Local Therapists	No	-	-	No	Yes
Self-Help Features					
Mood Tracking	Basic	Yes	Yes	Basic	Yes
Guided Activities	No	Yes	Yes	Limited	Yes
Gamified Exercises	No	Some	Some	No	Yes
Assessment Tools	Basic	No	No	No	Yes

Figure 3: Comparison of Mental Health Apps Available in Nepal

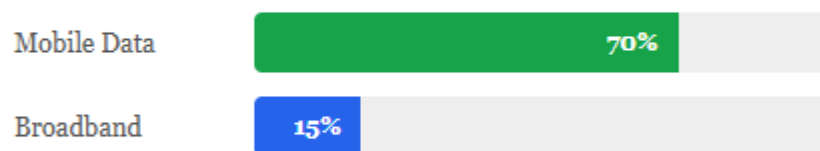
Local attempts to fill this gap have had limited success. A review of South Asian mental health apps by (Chandrashekar, 2018) found that most offer basic features like stress tips or breathing exercises but lack any connection to real therapists. Some Nepali organizations have created mental health content in local languages, yet these remain mostly informational rather than interactive. The few apps that do exist often copy Western models without considering that a farmer in Terai might need different

support than a student in Boston. Importantly, none of the available local options work properly offline, which matters when only 70 percent of Nepalis have regular internet access (Nepal Telecommunications Authority, 2022).

Internet Availability by Region



Type of Internet Access (National)



Key Challenge:

58% of rural Nepal lacks reliable internet - most international mental health apps require constant connectivity to function

Figure 4: Internet Access Challenges for Digital Mental Health in Nepal (Source: Nepal Telecommunications Authority)

[For detailed app comparison please see [Appendix C: Detailed App Comparison](#)]

1.4. Project as a Solution

Resilio integrates what other apps separate: therapist booking, live video/audio sessions, mood tracking, validated questionnaires, and culturally relevant content in one platform. The app includes gamified exercises like breathing games and wellness trivia, daily motivational quotes, and educational videos from local therapists. These features are proven to improve retention in Asian contexts (Arjadi et al., 2015; Fleming et al., 2019; Naeem et al., 2021). Offline functionality ensures rural users can access core features without internet. Payment through eSewa and affordable local pricing makes it accessible to all economic backgrounds. By combining professional support with self-help tools in a single platform designed for Nepal's reality, Resilio addresses the gaps left by existing solutions.

[For detailed feature breakdown please see [Appendix D: Detailed Solution Features](#)]

2. Aim and Objectives

2.1. Aim

The main aim of this project is to develop Resilio, a cross-platform wellness application providing personalized mental health support, interactive wellness content, and secure access to licensed therapists, following user-centred design principles and evidence-based intervention strategies.

2.2. Objectives

- Develop a cross-platform mobile application using Flutter and Dart with Node.js backend ensuring code maintainability.
- Implement offline functionality using SQLite, LRU caching, and Protocol Buffer serialization for uninterrupted access.
- Create mood tracking and lifestyle logging features based on ecological momentary assessment principles.
- Develop multimedia players supporting audio, video, and images aligned with relaxation response research.
- Build interactive wellness games including breathing exercises, trivia, affirmation builders, and mental health role-playing games leveraging gamification.
- Implement therapist booking with questionnaire-based recommendations and secure WebRTC-based video/audio sessions.
- Integrate Esewa payment gateway following secure transaction protocols.
- Create administrative dashboards for content, user, and therapist management.
- Ensure compliance with data protection standards and accessibility guidelines.

3. Expected Outcome & Deliverables

The project aims to deliver a complete mental wellness application with stable performance, secure data handling, and tools that support both users and therapists. The system focuses on accessible care, guided activities, and structured support. The expected outcome is a functional mobile app, a backend service, admin panel that together create a reliable environment for counselling, progress tracking, and content management.

3.1. Features

Table 1: Features

Features	Descriptions	Purpose
Authentication	Email/password, Google, Facebook, Supertokens (Password Less Login)	Secure and flexible user access
Therapist Booking	Schedule sessions with qualified therapists	Provides professional support
Live Sessions	Video or audio calls with therapists	Real-time guidance and counselling
Questionnaire & Recommendations	User fills forms; app suggests suitable therapists	Personalized care
Mood Tracking	Daily logging of mood and activities	Monitor mental health trends
Motivational Content	Quotes, tips, short reels, images	Enhance motivation and positivity
Multimedia Playback	Therapist videos, calming sounds, audio guides	Support self-help and relaxation
Games & Exercises	Breathing exercises, trivia, affirmation builder, role-playing game	Encourage engagement and coping skills
Offline Support	Local caching with SQLite, LRU, protocol buffers	Ensure accessibility without internet

Payment Integration	Esewa for subscription and therapy fees	Secure and localized transactions
Notifications	Reminders for sessions, tasks, and updates	Maintain engagement
User Profile & Settings	Manage account, privacy, theme, and activity	Personalization and control
Admin Dashboard	Manage content, users, therapists, and analytics	Platform administration

3.2. Deliverables

Table 2: Deliverables

Submission Type	Description
Project Proposal	Comprehensive proposal document with methodology, WBS, timeline, and feature specifications
Development Artefacts	Source code, database schemas, API documentation, offline caching implementation details
Application Code	Flutter mobile application, Node.js backend, web admin dashboard with all integrated features
Documentation	Technical documentation, user manuals, deployment guides, prototype iteration reports
Meeting Logs	Weekly progress logs, supervisor meeting minutes, decision records, stakeholder feedback summaries
Client Letter	Formal project acceptance and completion confirmation letter
Testing Reports	Unit, integration, system, performance, security, and user acceptance testing results
Multimedia & Content	Motivational quotes, tips, reels, therapist videos, audio files, and game assets
Analytics & Admin Data	Reports from admin dashboard including user engagement, subscription stats, and session logs

4. Project Risks, Threats and Contingency Plans

Table 3: Project Risks, Threats and Contingency Plans

Risk ID	Risk Description	Impact (1-5)	Likelihood (1-5)	Risk level (Impact x Likelihood)	Mitigation Strategy	Contingency Plan
R1	Data privacy breach exposing sensitive user information	3	4	12	Implement AES-256 encryption, OAuth 2.0 authentication, regular security audits	Immediate patch, user notification, temporary service lockdown
R2	Low user engagement reducing application effectiveness	3	3	9	Incorporate gamification elements, push notifications, personalized content recommendations	Collect feedback, adjust content and features to improve retention
R3	WebRTC session failures during therapist consultations	3	4	12	Implement audio-only fallback, connection quality monitoring, session reconnection mechanisms	Switch to audio sessions, reschedule failed sessions

R4	Esewa payment integration failures	2	4	8	Comprehensive SDK testing, transaction logging, manual payment fallback options	Manual payment processing, alternative payment link
R5	Offline synchronisation conflicts causing data inconsistency	3	3	9	Implement conflict resolution algorithms, timestamp-based merging, user notification for conflicts	Manual sync, notify user of conflicts
R6	Scope creep extending development timeline	3	3	9	Strict change control process, prioritized feature backlog, regular stakeholder communication	Reprioritize tasks, defer non-critical features
R7	Third-party API changes breaking functionality	2	3	6	Abstract API dependencies, version locking, monitoring for deprecation notices	Update integration, revert to previous stable version if needed
R8	Therapist no-shows affecting user trust	2	4	8	Automated reminders, cancellation policies, backup therapist recommendations	Notify users, reschedule session, assign backup therapist

R9	OOM error in short reels causing app crash	3	3	9	Optimize video loading, implement caching, limit concurrent media processing	Restart service, limit reel length, notify users
R10	Unexpected backend downtime affecting all services	4	3	12	Use load balancers, implement server redundancy, monitor system health	Failover to backup server, alert team, restore from backups

Risk Level Indicator:Extreme Risk: High Risk: Medium Risk: Low Risk: 

5. Methodology

5.1. SDLC Models Comparison

Selecting the right development approach is critical for project success. Three models were evaluated based on their fit with Resilio's requirements, academic constraints, and the need for user-centred mental health features.

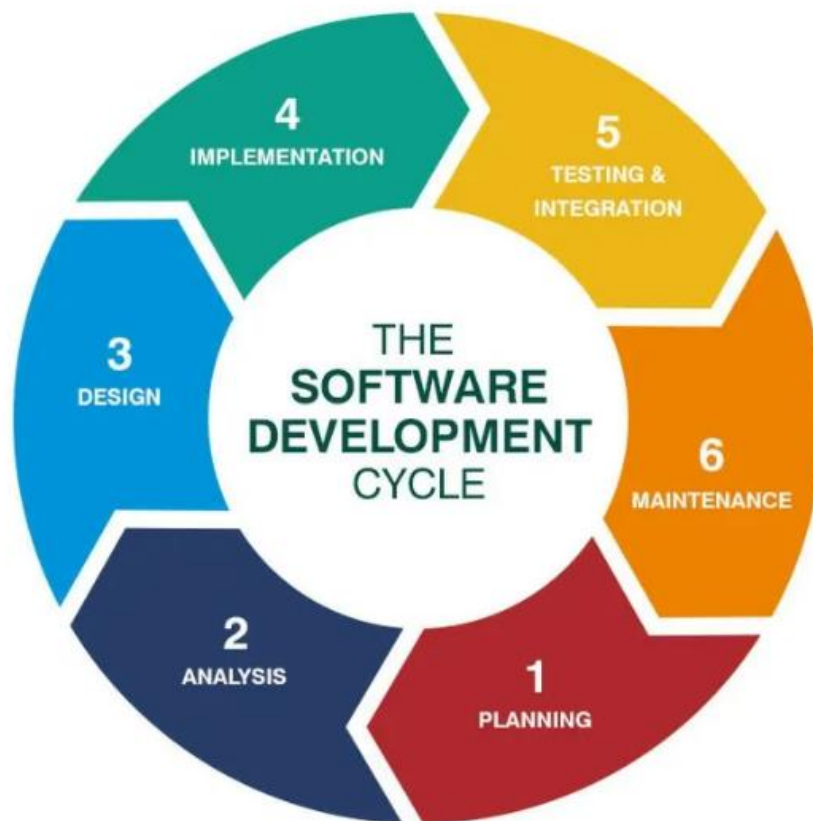


Figure 5: Software Development Lifecycle

1. Agile Model:

Agile emphasizes iterative development, frequent stakeholder feedback, and rapid adaptation to change (Beck et al., 2001). Teams deliver working software in short sprints. However, Agile struggles with scope creep and light documentation. For academic projects with fixed deadlines, aligning sprints with submission milestones is challenging.



Figure 6: Agile Model

2. Rational Unified Process (RUP):

RUP is an iterative, use-case-driven model divided into four phases: Inception, Elaboration, Construction, and Transition (Kruchten, 2003). It balances structured documentation with iterative development. Risk management is built into early phases, ensuring technical challenges are addressed before major construction. Stakeholders provide input at defined points throughout.

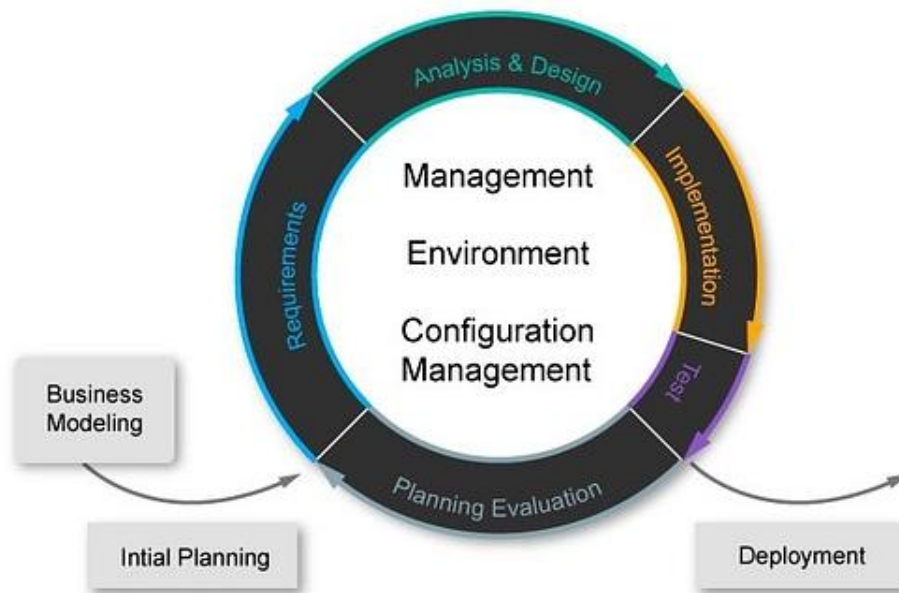


Figure 7: RUP Model

3. Prototype Model:

This approach focuses on building early prototypes for user evaluation (Sommerville, 2016). It supports rapid visualization and early feedback. However, extensive prototyping may increase effort without clear boundaries, risking incomplete documentation for academic deliverables.

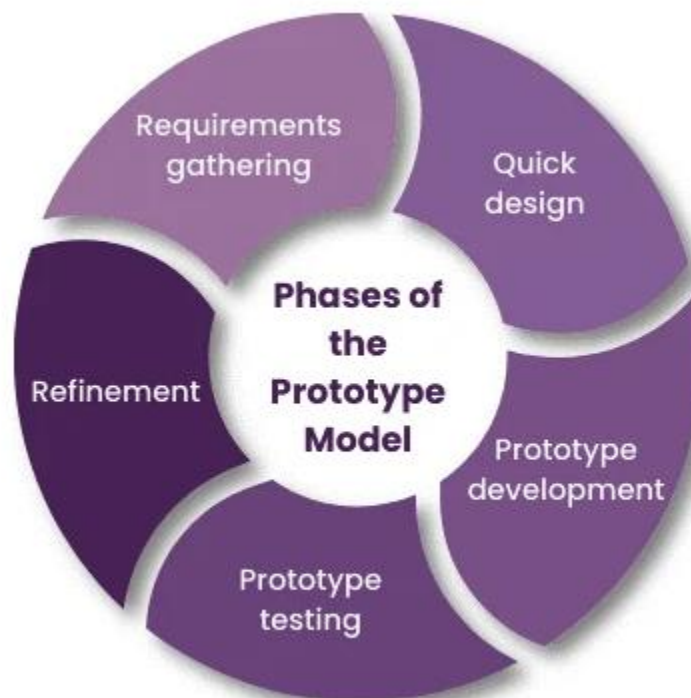


Figure 8: Prototype Model

[For detailed model comparison matrix please see [Appendix E: Methodology Details](#)]

5.2. Chosen Methodology: Rational Unified Process (RUP)

RUP was selected for Resilio's development based on several factors aligned with project needs.

- **Iterative Development with Structure:**

RUP allows phased development with iterative cycles, supporting continuous refinement of mood tracking and therapist sessions while maintaining clear milestones.

- **Requirement Clarity:**

RUP's Elaboration phase addresses requirement refinement through stakeholder review, ensuring features meet actual user needs.

- **Risk Management:**

Technical challenges like WebRTC integration and offline storage are addressed during Elaboration, validating architecture before major construction (Kruchten, 2003).

- **Stakeholder Engagement:**

Each phase includes defined points for user evaluation and expert review, ensuring clinical appropriateness.

- **Alignment with Academic Timeline:**

RUP phases map naturally to academic milestones—Inception to proposal, Elaboration to mid-term, Construction to development, and Transition to final submission.

6. Resources Requirements

6.1. Hardware Requirements

- **Development Computer:**
MacBook Air M4 (Apple Silicon) with 16GB RAM and 256GB SSD for application development and testing.
- **Android Device:**
Android smartphone running Android 8.0 or higher with minimum 4GB RAM for Android application testing.
- **iOS Device:**
iPhone 8 or newer running iOS 13+ for iOS application testing.
- **Tablet Device:**
iPad or Android tablet for responsive design testing.
- **Webcam and Microphone:**
Built-in HD FaceTime camera and microphone for WebRTC feature testing.

6.2. Software Requirements

Table 4: Software Requirements

Category	Description
Frontend Development	Flutter SDK and Dart for UI and logic. Android Studio and Xcode for platform builds.
Backend Development	Node.js and Express.js for routing, sessions, security checks, and API layers. npm for package management.
Database	SQLite for offline data. PostgreSQL for server storage. Firebase for authentication.
Cloud Services	Supabase for database hosting and authentication flow. Cloudinary for media storage. Firebase Cloud Messaging for push alerts.
Real-time Communication	WebRTC for live sessions. Agora SDK for stable calls and fallback controls.
Payment Integration	eSewa SDK for secure mobile payments and subscription unlock.
Version Control	Git for tracking changes. GitHub for repository hosting and pull requests.
Design Tools	Figma for UI layouts. Adobe XD for flow mapping and prototypes.
Testing Tools	Flutter Test for widget and unit tests. Postman for API checks. Jest for backend tests.
Development IDE	Visual Studio Code for backend and scripts. Android Studio and Xcode for builds and debugging.

7. Work Breakdown Structure

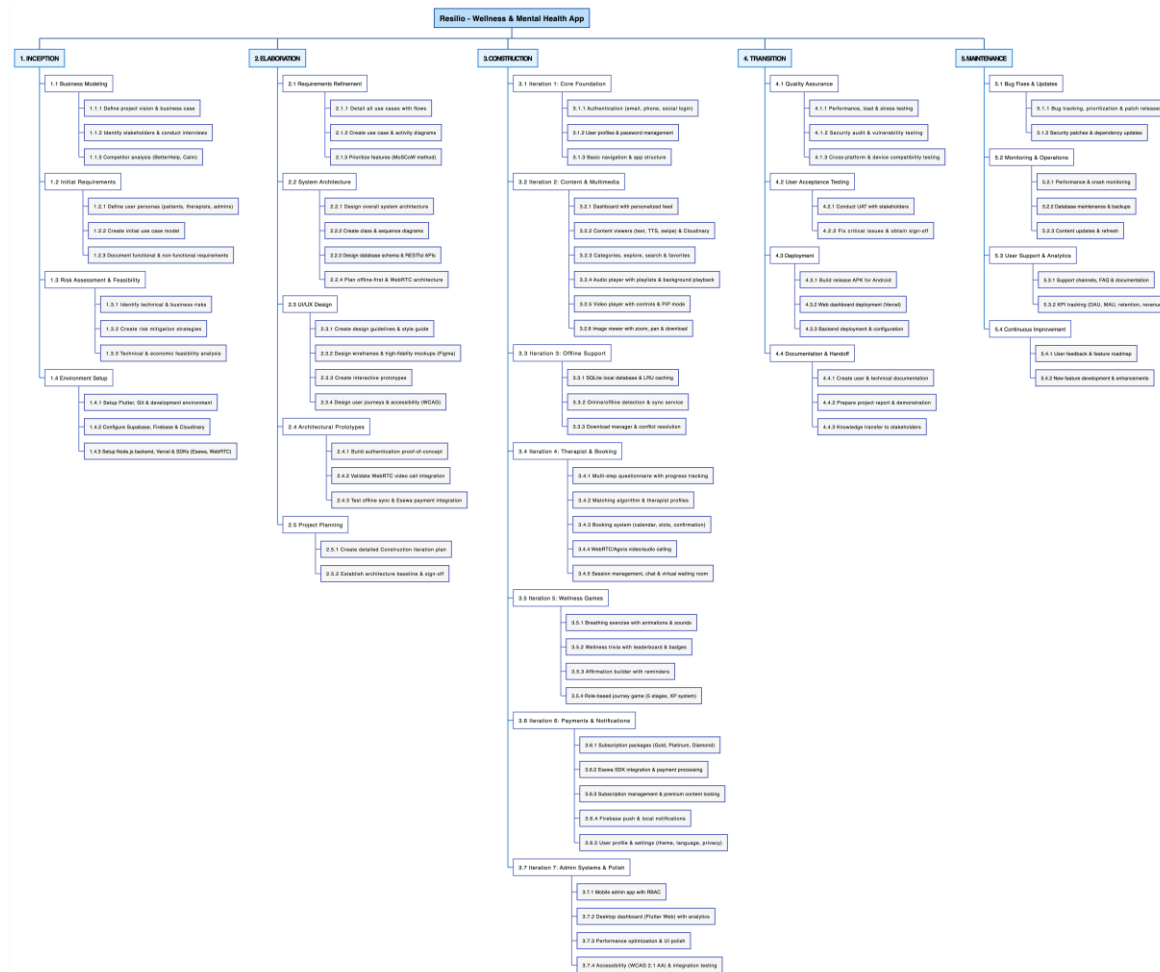


Figure 9: Work Breakdown Structure

[For a clear view of WBS, please [Click Here](#)]

8. Milestones



Figure 10: Milestone Timeline Chart

9. Project Gantt Chart

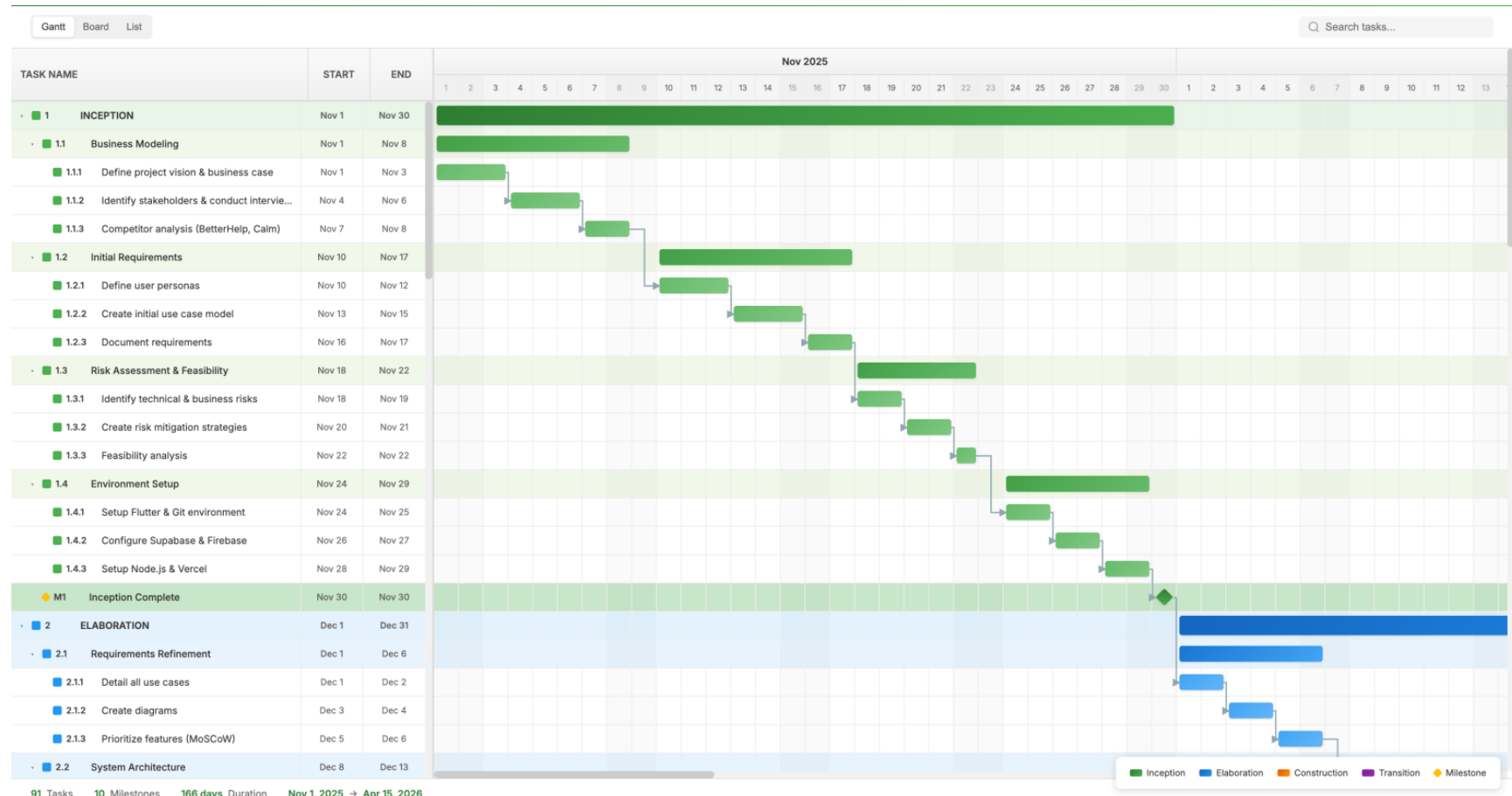


Figure 11: Gantt Chart -1

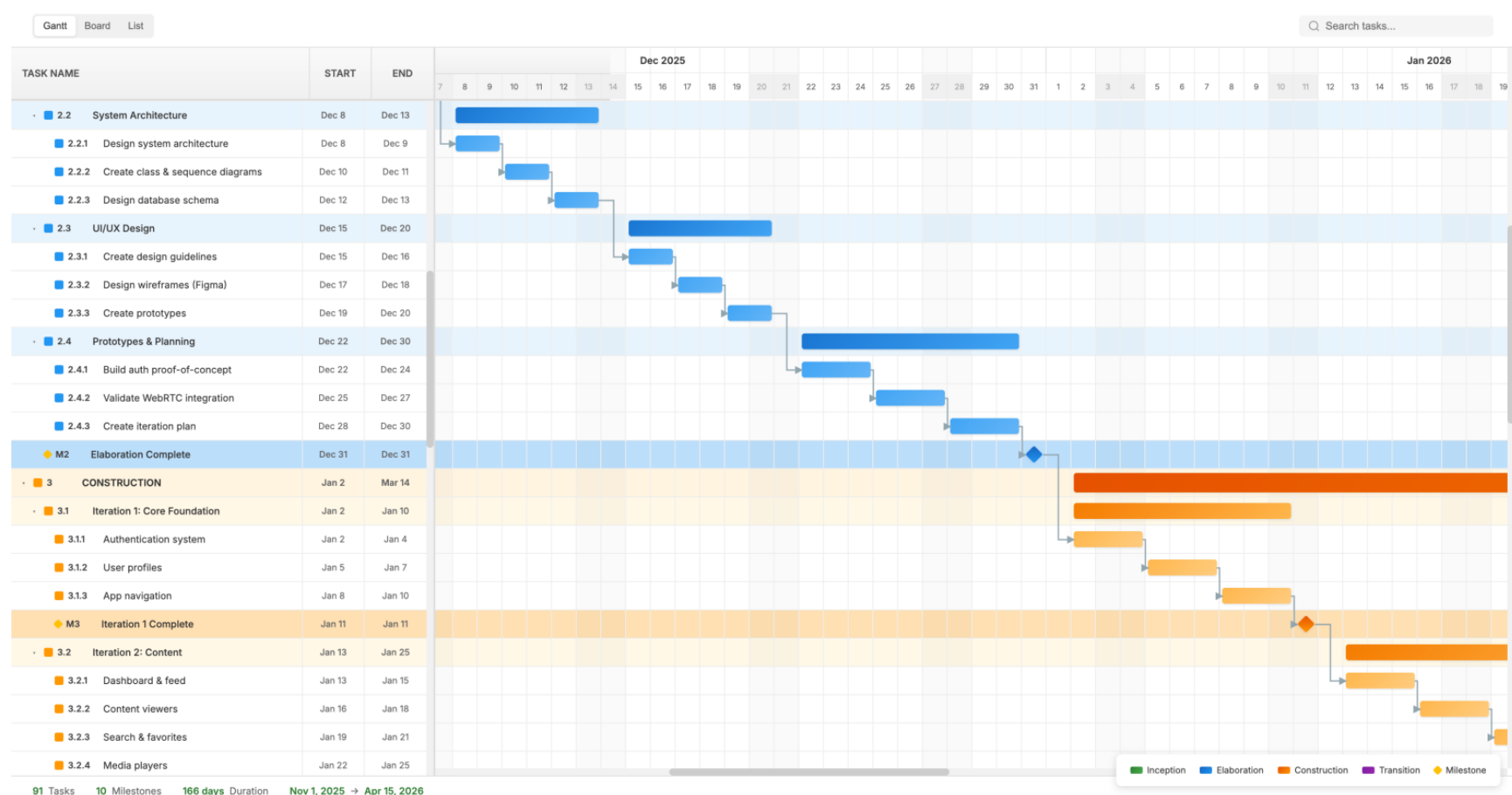


Figure 12: Gantt Chart -2

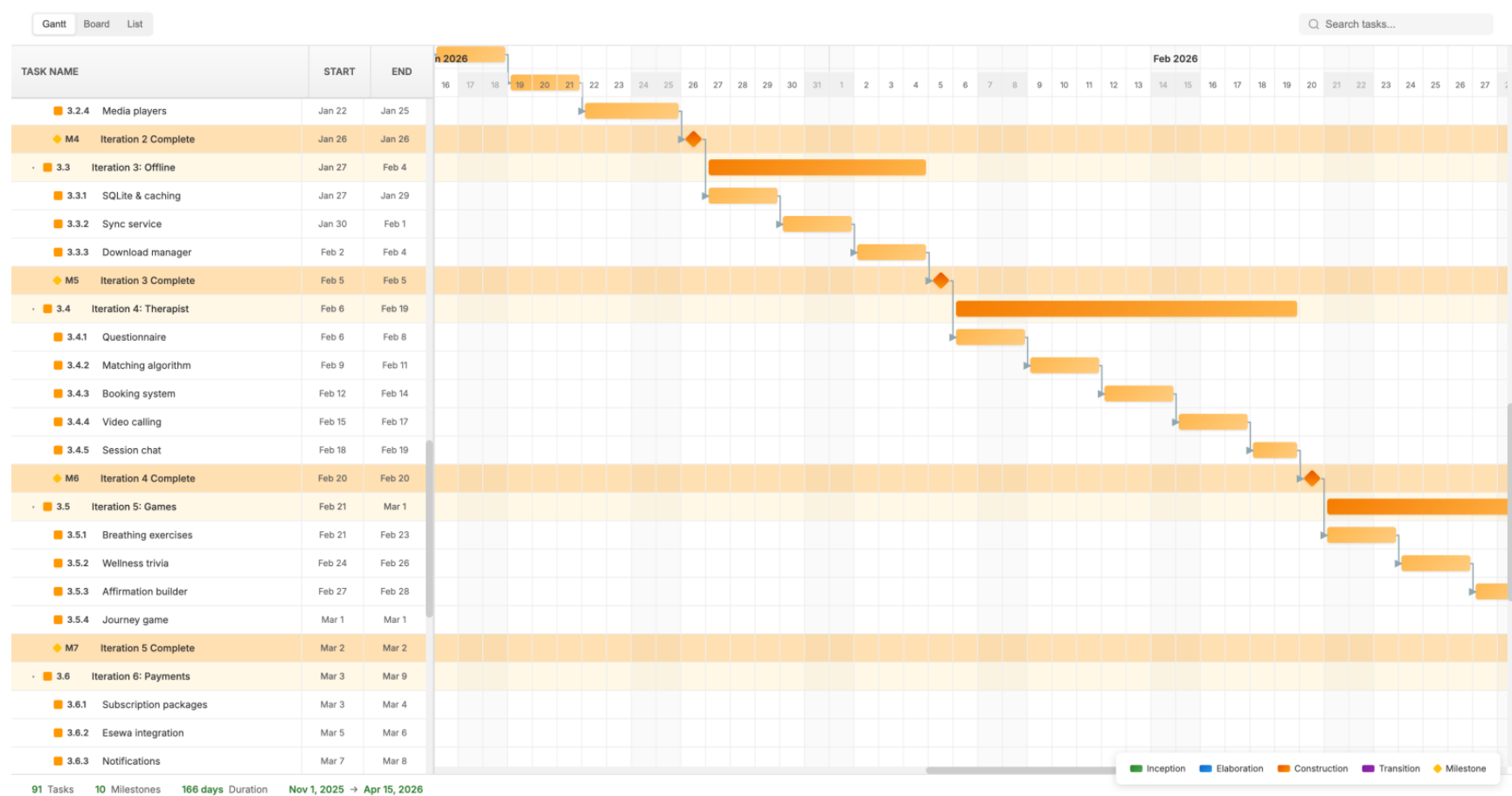


Figure 13: Gantt Chart -3

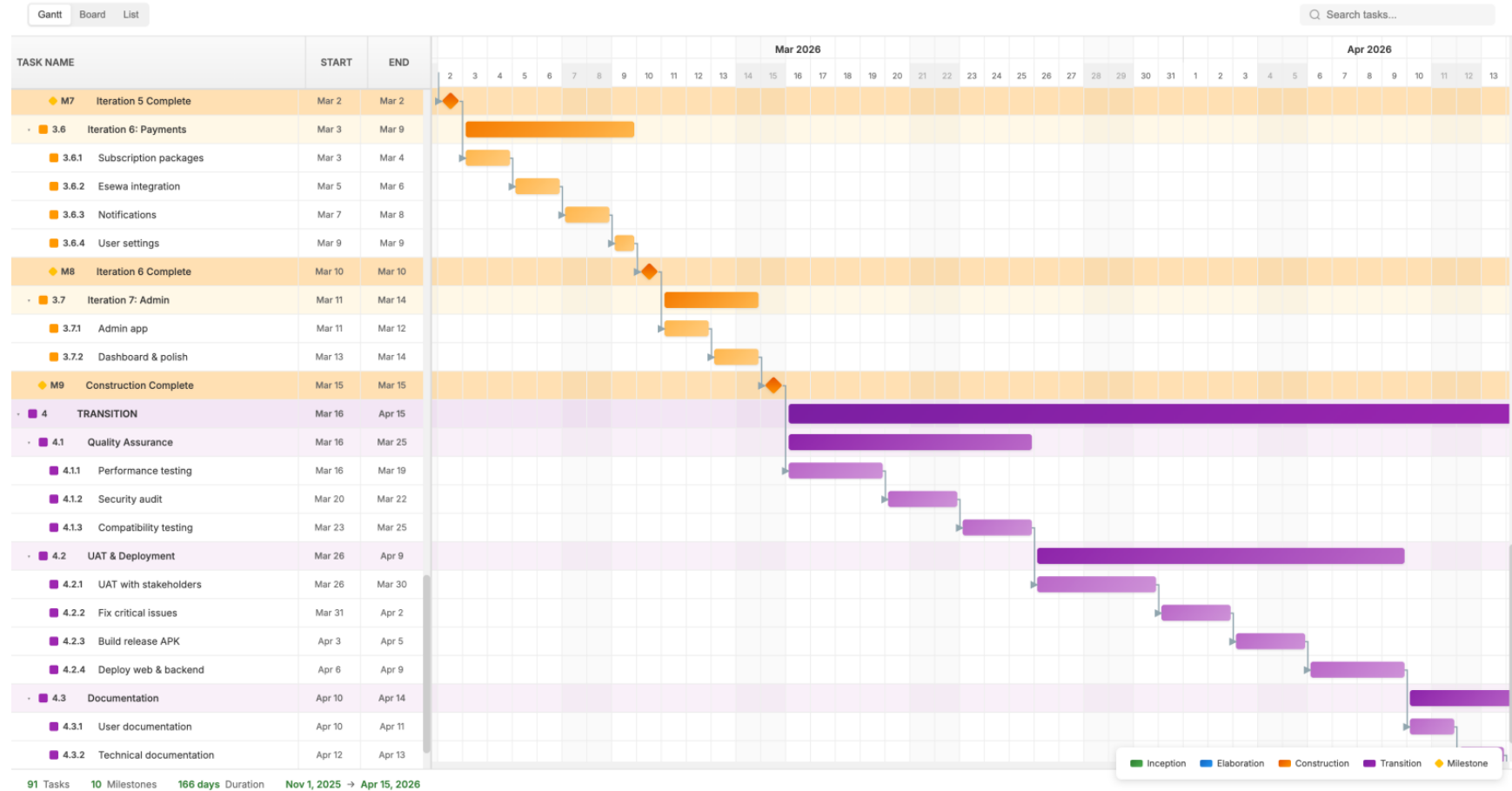


Figure 14: Gantt Chart -4

10. Conclusion

Resilio addresses critical gaps in mental health support accessibility by integrating mood tracking, multimedia content, gamified wellness activities, and secure teletherapy within a single platform. The offline-first architecture ensures functionality regardless of connectivity constraints, particularly relevant for the Nepali context. Adopting the Prototype Model enables iterative refinement based on user and expert feedback, ensuring the application meets genuine user needs. Through comprehensive feature implementation, robust security measures, and evidence-based design principles, Resilio demonstrates a structured approach to developing holistic mental health management solutions.

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12. Appendix

Appendix A: Detailed Mental Health Statistics in Nepal

[Return to Section [1.1. Background and Context](#)]

The bigger problem is that almost nobody gets treatment. (Luitel et al. 2015) estimated that over 90 percent of people with mental health issues in Nepal never receive any professional care. There are many reasons for this. The country has very few mental health workers, less than one psychiatrist for every 200,000 people according to WHO data (World Health Organization, 2017). Most services are in Kathmandu and other cities while rural communities have almost nothing. On top of that, many people feel ashamed to talk about mental health or see it as a spiritual matter rather than a medical one (Kohrt and Harper, 2008). Cost and travel also stop people from seeking help.

The pandemic made things more urgent. WHO reported a 25 percent global increase in anxiety and depression during 2020, and Nepal saw similar patterns with increased calls to suicide helplines and mental health services (Sigdel et al., 2021). Studies now show that video therapy can be just as effective as meeting in person (Backhaus et al., 2012), which should make digital solutions more viable. But you still need apps that actually work in local contexts. Right now, if someone in rural Nepal wants mental health support through their phone, they have to choose between expensive foreign apps they cannot afford, basic local apps that offer little real help, or trying to piece together different tools that were never meant to work together.

A.1 Prevalence of Mental Disorders

The Nepal Mental Health Survey conducted in 2020 provides the most comprehensive data on mental health conditions in the country. The survey covered all seven provinces and included both urban and rural populations.

Condition	Lifetime Prevalence (%)	12-Month Prevalence (%)
Any Mental Disorder	12.9	5.8
Depression	4.5	2.1
Anxiety Disorders	3.5	1.8
Alcohol Use Disorder	4.0	2.5
Suicidal Ideation	5.9	2.3
Suicide Attempts	1.4	0.5

Source: Ministry of Health and Population (2020)

These numbers translate to approximately 3.5 million Nepali adults who have experienced some form of mental health condition. Depression and anxiety together affect around 2.3 million people. The suicidal ideation rate of 5.9 percent means roughly 1.7 million adults have thought about ending their lives at some point.

A.2 Treatment Gap Analysis

The treatment gap refers to the percentage of people who need mental health care but do not receive it. Nepal has one of the highest treatment gaps in the world.

Indicator	Nepal	Low-Income Countries Average	High-Income Countries Average
Treatment Gap (Overall)	92%	85%	45%
Treatment Gap (Depression)	89%	80%	50%
Treatment Gap (Severe Mental Disorders)	75%	70%	35%

Source: Luitel et al. (2015); Patel et al. (2018)

This means that for every 100 people with mental health problems in Nepal, only 8 receive any form of professional care. The remaining 92 either cope on their own, rely on traditional healers, or suffer without any support.

A.3 Mental Health Workforce

Nepal has a severe shortage of trained mental health professionals. The following table compares Nepal's mental health workforce with WHO recommendations.

Professional Type	Nepal (per 100,000)	WHO Recommended (per 100,000)	Gap
Psychiatrists	0.36	1.0	64% shortage
Psychologists	0.08	1.0	92% shortage
Mental Health Nurses	0.25	2.0	88% shortage
Social Workers	0.05	1.0	95% shortage

Source: World Health Organization (2017)

With approximately 100 psychiatrists serving a population of 30 million, each psychiatrist would need to care for 300,000 people. Most of these professionals are based in Kathmandu, leaving rural areas with almost no coverage.

A.4 Geographic Distribution of Services

Mental health services in Nepal are heavily concentrated in urban areas:

Region	% of Population	% of Mental Health Services
Kathmandu Valley	12%	65%
Other Urban Areas	18%	25%
Rural Areas	17%	10%

Source: Luitel et al. (2015)

This means 70 percent of the population has access to only 10 percent of mental health services. A person in rural Humla or Dolpa may need to travel 2-3 days to reach the nearest psychiatrist.

A.5 Barriers to Treatment

Research has identified several key barriers preventing Nepalis from seeking mental health care:

Barrier	% Reporting This Barrier
Cost of treatment	78%
Distance to services	65%
Stigma and shame	62%
Belief it is not a medical issue	48%
Lack of awareness about services	45%
Preference for traditional healers	38%

Source: Kohrt and Harper (2008); Luitel et al. (2015)

Cost and distance are the top barriers, followed closely by stigma. Many Nepalis still view mental illness as a result of karma, spiritual possession, or personal weakness rather than a treatable medical condition.

[Return to Section [1.1. Background and Context](#)]

Appendix B: Extended Problem Analysis

[Return to Section [1.2. Problem Domain](#)]

B.1 App Fragmentation Problem

The current mental health app landscape is highly fragmented. Users who want comprehensive support must use multiple disconnected applications:

Scenario 1: User seeking therapy and self-help

- Downloads BetterHelp for therapist access
- Downloads Daylio for mood tracking
- Downloads Calm for meditation
- Downloads Youper for anxiety exercises

Result: 4 apps that do not share data or work together

Scenario 2: User in Nepal seeking affordable support

- Tries BetterHelp but cannot afford \$70/week
- Downloads free meditation app but content is in English
- Finds Nepali mental health website but it is only informational

Result: No integrated solution available

Bakker et al. (2016) studied 2,000 mental health app users and found:

- Average user downloads 3.2 mental health apps
- 74% stop using apps within 2 weeks
- Main reason for abandonment: “felt disconnected from real care”
- Users who had therapist integration showed 3x higher retention

B.2 Lack of Integration with Professional Care

Most self-help apps operate completely separate from professional mental health services. This creates several problems:

Problem	Impact
Therapists cannot see client mood data	Treatment decisions based on incomplete information
Self-assessments not shared with professionals	Duplicate testing, wasted time
No continuity between sessions	Progress lost; users disengage
Crisis situations not flagged	Delayed intervention in emergencies
Self-help exercises not reinforced	Users unsure if doing exercises correctly

(Patel et al. 2011) argued that low-resource countries especially need integrated models because they cannot afford to waste limited professional time on tasks that apps could support.

B.3 Long-term Engagement Failures

(Torous et al. 2019) reviewed 1,000+ mental health apps and found critical engagement problems:

Metric	Finding
Apps with evidence-based content	Only 3%
Apps designed for long-term use	Only 8%
Apps with therapist integration	Only 5%
Apps working offline	Only 12%
Apps with gamification elements	Only 15%

Most apps are designed for short-term use or one-time downloads. They lack features that bring users back daily or that create meaningful progress over time.

B.4 Connectivity Requirements

Mental health apps typically require constant internet connectivity:

App	Offline Capability
BetterHelp	None - all features require internet
Calm	Partial - some downloaded meditations
Headspace	Partial - limited offline content
Wysa	Minimal - basic chatbot only
Woebot	None - requires constant connection

For Nepal where 30% of the population lacks regular internet and many areas have unstable connectivity, these requirements make most apps unusable for rural populations.

B.5 User Journey Without Integrated Solution

A typical user journey in Nepal currently looks like:

1. **Awareness Stage:** Person realizes they might need help
2. **Search Stage:** Looks for apps, finds mostly English options
3. **Trial Stage:** Downloads 2-3 free apps
4. **Frustration Stage:** Apps feel disconnected, content irrelevant
5. **Abandonment Stage:** Stops using apps within 2 weeks
6. **Gap Stage:** Returns to no support or traditional healers
7. **Deterioration Stage:** Condition may worsen without intervention

An integrated solution like Resilio aims to break this cycle by providing everything in one place.

[Return to Section 1.2. Problem Domain]

Appendix C: Detailed App Comparison

[Return to Section [1.3. Current Scenario](#)]

C.1 International Apps Analysis

BetterHelp:

Feature	Details
Founded	2013, USA
Primary Service	Online therapy with licensed counsellors
Cost	\$60-90 per week (\$240-360/month)
Languages	English only
Therapist Location	Mostly US, UK, Canada
Offline Mode	Not available
Payment Methods	Credit card, PayPal only
Nepal Accessibility	Very low - cost prohibitive, no local therapists

Headspace:

Feature	Details
Founded	2010, UK/USA
Primary Service	Meditation, mindfulness courses
Cost	\$69.99/year or \$12.99/month
Languages	English, some dubbed content
Professional Support	None - self-help only
Offline Mode	Limited downloaded sessions
Payment Methods	Credit card, mobile payments
Nepal Accessibility	Low - similar barriers as Calm

C.2 Local and Regional Apps

Available Nepali Mental Health Resources:

Resource	Type	Limitations
CMC Nepal Helpline	Phone hotline	Not an app, limited hours
TPO Nepal Resources	Website/PDF	Informational only, not interactive
Various meditation apps	Mobile apps	Basic features, no therapist integration
Facebook support groups	Social media	Privacy concerns, no professional oversight

Most local options are either phone-based hotlines or static informational websites. Interactive, integrated mobile apps designed for Nepal remain largely unavailable.

Appendix D: Detailed Solution Features

[Return to Section [1.4. Project as a Solution](#)]

D.1 Core Feature Breakdown

Therapist Booking System:

Feature	Description
Therapist Profiles	Detailed profiles with qualifications, specializations, languages
Availability Calendar	Real-time availability with time zone support
Session Types	Video call, audio call, chat-based options
Booking Management	Reschedule, cancel, receive reminders
Therapist Matching	Recommendations based on user needs and preferences

Live Session Capabilities:

Feature	Description
Video Sessions	HD video with adaptive quality for low bandwidth
Audio Sessions	High-quality audio with minimal data usage
Session Recording	Optional recording for user review (with consent)
Notes Integration	Therapist can access user mood data during session
Crisis Protocol	Emergency procedures if user shows distress signs

Mood Tracking System:

Feature	Description
Daily Check-ins	Quick mood logging with emoji and scale options
Mood History	Visual graphs showing patterns over time
Trigger Tracking	Log events, activities, or thoughts affecting mood
Insights	AI-generated observations about mood patterns
Therapist Sharing	Option to share mood data with assigned therapist

Mental Health Questionnaires:

Questionnaire	Purpose	Frequency
PHQ-9	Depression screening	Weekly/Monthly
GAD-7	Anxiety screening	Weekly/Monthly
WHO-5	Wellbeing index	Weekly
Custom Assessments	Specific concerns	As needed

All questionnaires are validated instruments adapted for Nepali context.

D.2 Engagement Features

Gamified Exercises:

Game	Mental Health Benefit	How It Works
Breathing Bubble	Anxiety reduction, relaxation	Follow expanding/contracting bubble to regulate breath
Wellness Trivia	Mental health education	Answer questions about mental health facts
Affirmation Builder	Self-esteem, positive thinking	Create and customize personal affirmations

Research shows gamification increases app retention by 40-60% in mental health contexts (Fleming et al., 2019).

Motivational Contents:

Content Type	Frequency	Personalization
Daily Quotes	Every morning	Based on current mood and goals
Wellness Tips	2-3 per day	Matched to user's focus areas
Short Videos	Updated weekly	Local therapists, Nepali context
Podcasts	Updated weekly	Mental health topics in Nepal
Success Stories	Updated monthly	Real recovery journeys (anonymized)

Research shows gamification increases app retention by 40-60% in mental health contexts (Fleming et al., 2019).

Progress and Rewards:

Feature	Description
Streak Tracking	Consecutive days of app usage
Achievement Badges	Milestones for exercises, sessions, mood logs
Progress Reports	Weekly/monthly summaries of improvement
Goal Setting	Set and track personal mental health goals

D.3 Offline Functionality**How Offline Mode Works:**

Component	Offline Capability
Mood Tracking	Full functionality, syncs when online
Saved Exercises	All downloaded content accessible
Questionnaires	Can complete offline, submits when connected
Motivational Content	Cached quotes and tips available
Session Notes	View past session summaries
Progress Data	Full access to historical data

Technical Implementation:

Technology	Purpose
SQLite Database	Local storage of user data
LRU Caching	Efficient content storage
Protocol Buffers	Compressed data for fast sync
Background Sync	Automatic upload when connected
Conflict Resolution	Handles data conflicts gracefully

Data Usage Optimization:

Activity	Purpose
Daily sync	~2-5 MB
Video session (30 min)	~150-300 MB
Audio session (30 min)	~30-50 MB
Content download	~10-20 MB weekly

Users can pre-download content on WIFI and use app features on minimal data.

D.4 Payment Integration**eSewa Integration:**

Feature	Description
Direct Payment	Pay from eSewa wallet
Subscription	Monthly/yearly plans
Session Payment	Pay per therapy session
Content Unlock	Purchase premium content
Refunds	Processed through eSewa

D.5 Cultural Adaptation

Nepali Context Features:

Feature	Description
Language	Full Nepali language support
Cultural Content	Quotes from Nepali literature, local wisdom
Local Therapists	Nepali-speaking, culturally trained professionals
Stigma-Aware Design	Private, discreet app appearance
Family Dynamics	Content addressing family-related stress

Addressing Local Barriers:

Barrier	How Resilio Addresses It
Cost	Affordable pricing, free tier available
Distance	Remote therapy, no travel needed
Stigma	Private app, discreet notifications
Language	Full Nepali support
Internet	Offline functionality
Trust	Local therapists, familiar payment methods

D.6 User Journey with Resilio

Week 1: Onboarding

- Download app, create account
- Complete initial assessment questionnaires
- Set mental health goals
- Explore free content and exercises
- Start daily mood tracking

Week 2-4: Building Habits

- Daily mood logs become routine
- Complete gamified exercises
- Read daily motivational content
- Patterns begin emerging in mood data
- Consider booking first therapy session

Month 2: Integrated Care

- Regular therapy sessions (if chosen)
- Therapist reviews mood data before sessions
- Personalized recommendations based on progress
- Engagement with community features
- Noticeable improvement in tracking

Ongoing: Continuous Support

- Maintenance therapy as needed
- Self-monitoring continues
- Access to new content and features
- Crisis support available if needed
- Long-term mental wellness management

[Return to Section [1.4. Project as a Solution](#)]

Appendix E: Methodology Details

[Return to Section [5. Methodology](#)]

E.1 Detailed SDLC Model Comparison

E.1.1 Agile Model Analysis

Agile emerged from the software industry's frustration with heavyweight processes that could not adapt to changing requirements. The Agile Manifesto, published in 2001, prioritized individuals over processes, working software over documentation, customer collaboration over contracts, and responding to change over following plans (Beck et al., 2001).

Strengths for Resilio:

Strength	Application
Rapid feedback	Users could test mood tracking early
Flexibility	Requirements could evolve based on therapist input
Continuous delivery	Features released incrementally
Team collaboration	Close coordination between developer and client

Weakness for Resilio:

Weakness	Impact
Light documentation	Academic requirements need detailed reports
Scope creep risk	Features could expand beyond timeline
Sprint alignment	Two-week sprints may not match submission dates
Product owner need	Requires dedicated decision-maker unavailable in academic context

Verdict: Agile's flexibility is attractive but its informal documentation and open-ended nature create risks for academic projects with fixed deliverables.

E.1.2 Waterfall Model Analysis

Though not selected for final comparison, Waterfall deserves mention as the traditional alternative. Royce (1970) originally described this sequential approach where each phase completes before the next begins.

Strengths for Resilio:

Strength	Application
Clear phases	Easy progress tracking
Heavy documentation	Meets academic requirements
Predictable timeline	Fixed deadlines manageable
Simple management	Linear progression easy to supervise

Weakness for Resilio:

Weakness	Impact
No iteration	Cannot refine features based on feedback
Late testing	Problems discovered only at end
Rigid requirements	Cannot adapt to user insights
High risk	Major issues surface late in project

Verdict: Waterfall's rigidity makes it unsuitable for user-centred mental health applications where feedback shapes design.

E.1.3 Prototype Model Analysis

Prototyping focuses on building quick versions for user evaluation. Sommerville (2016) describes two approaches: throwaway prototyping where prototypes are discarded after learning, and evolutionary prototyping where prototypes gradually become the final system.

Strengths for Resilio:

Strength	Application
User feedback	Test mood tracking interfaces early
Requirement clarity	Users show what they want, not just describe it
Risk reduction	Identify usability issues before full build
Stakeholder buy-in	Tangible demonstrations build confidence

Weakness for Resilio:

Weakness	Impact
Scope uncertainty	Endless refinement cycles possible
Documentation gaps	Focus on prototypes over reports
Effort duplication	May rebuild features multiple times
No clear phases	Hard to measure progress against deadlines

Verdict: Prototyping offers valuable user feedback but lacks structure for managing complete academic projects.

E.1.4 RUP Model Analysis

Kruchten (2003) developed RUP to combine the benefits of iterative development with the predictability of phased approaches. The model uses four phases containing multiple iterations.

Strengths for Resilio:

Strength	Application
Structured iteration	Refine features within defined phases
Use-case driven	Features trace to real user needs
Risk management	Address technical challenges early
Balanced documentation	Sufficient for academic needs
Stakeholder checkpoints	Regular expert review built in

Weakness for Resilio:

Weakness	Impact
Complexity	More overhead than pure Agile
Learning curve	Understanding RUP concept can be tough
Artifact heavy	Many documents to maintain

Verdict: RUP provides the structure academic projects need while allowing iteration that user-centred design requires.

E.1.5 Comparison Matrix

Criteria	Agile	Waterfall	Prototype	RUP
Flexibility	High	Low	High	Medium
Documentation	Light	Heavy	Light	Balanced
User Involvement	Continuous	End Only	Continuous	Phased
Risk Management	Reactive	Late	Reactive	Proactive
Structure	Low	High	Low	Medium
Iteration Support	High	None	High	Medium
Deadline Alignment	Difficult	Easy	Difficult	Easy
Academic Fit	Poor	Medium	Poor	Good

[Return to Section [5.1. SDLC Models Comparison](#)]

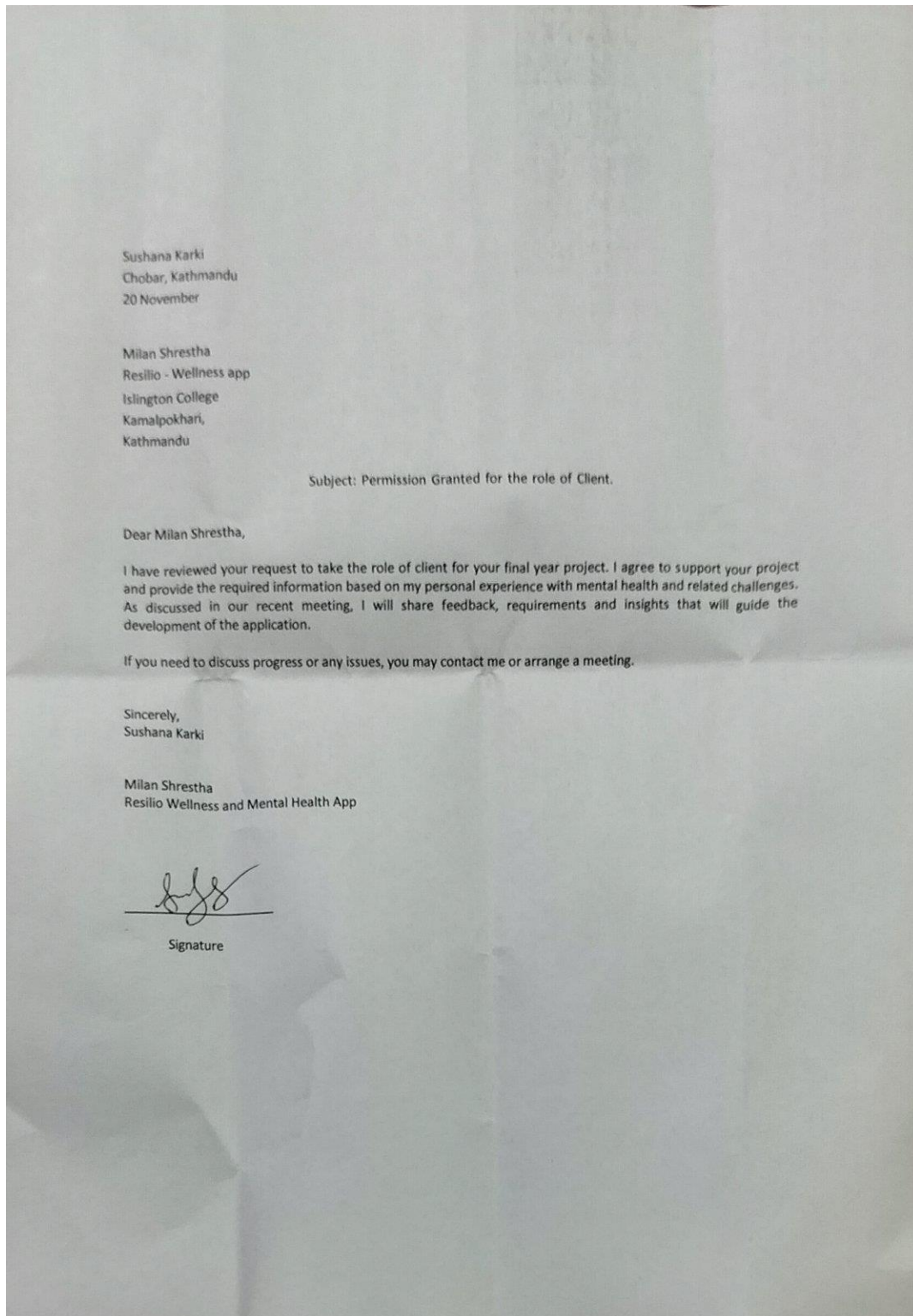
Appendix F: Client Letter

Figure 15: Client approval letter from Sushana Karki

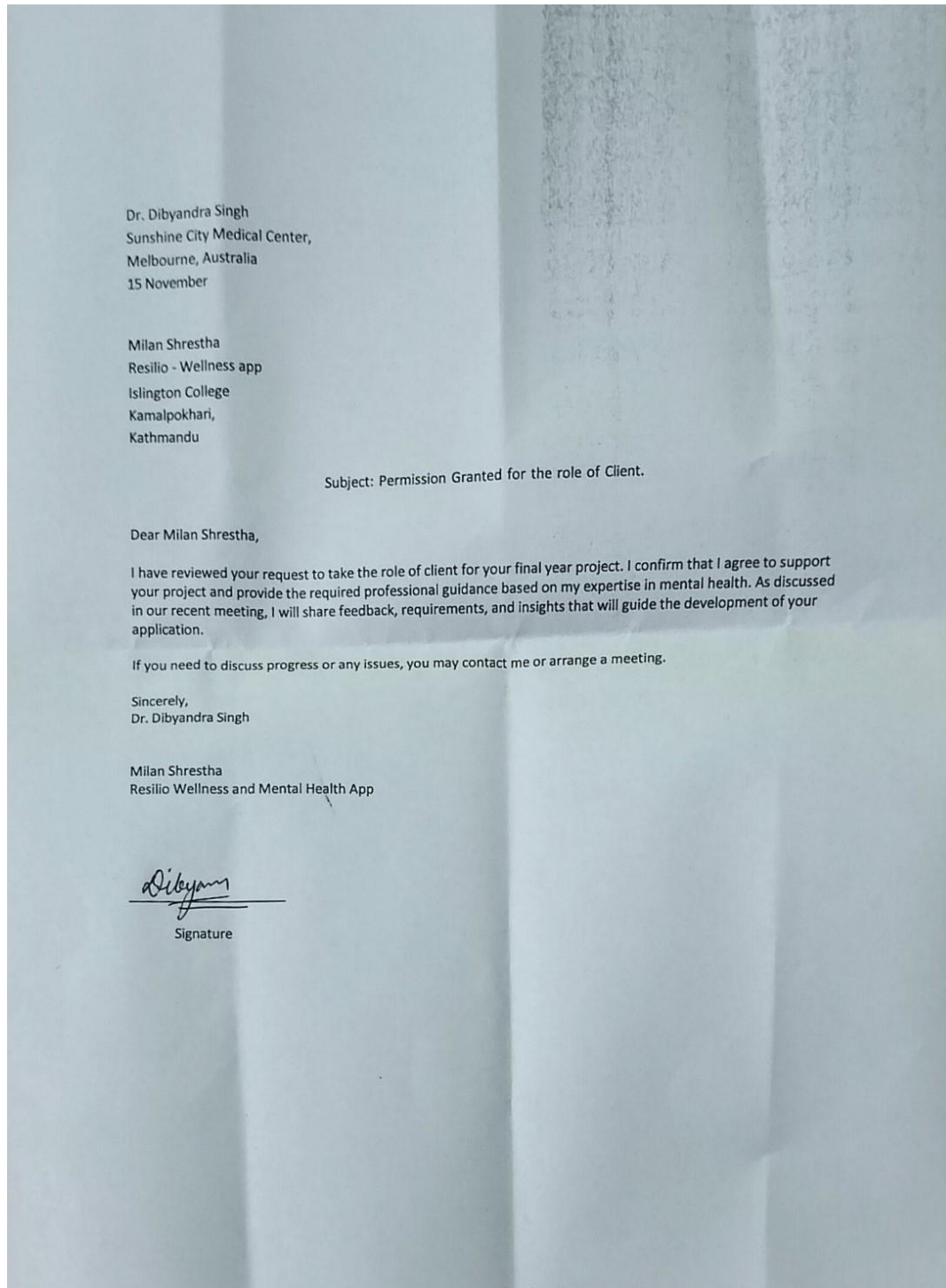


Figure 16: Client approval letter from Dr. Dibyandra Singh